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Perspectives of the degree project: Small-scale action research in Swedish higher education

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ABSTRACT

The degree project in educational sciences constitutes the pre-service teacher students' final academic achievement in the Bridging Teacher Education Programme at a university in Sweden. The degree project is complicated or a possibility because this university thesis constitutes only one part of the professional degree. The other part of the thesis project, a BA/BSc or MA/MSc, has been acknowledged as previous academic achievement, a required academic qualification for enrolment in the programme. A further difficulty or opportunity is that students' previous theses were achieved in academic disciplines not typically related to teacher education, sometimes several decades ago. This study aimed to visualise the notion of degree project in bridging teacher education. Our initial questions were: Could the transition between academic disciplines be facilitated? Could knowledge of degree project genres be visualised to enhance academic achievement? What is professional development for student teachers? In the actual study, the scope was limited to perspectives of the degree project and suggestions for improvement of degree project design. Thus, in participatory inquiry conversations, small groups of students and lecturers shared their perspectives of the degree project, and provided suggestions for improvement of the project design. Findings from qualitative analysis of the audio recorded data revealed students' difficulties in conceptualising the writing assignment, mainly due to unpreparedness. Means of addressing the unpreparedness emerged as task-sequencing throughout the study programme, applying an academic literacies approach. Implications for further action research and teaching are participatory inquiry conversations as continual components in the study programme. A further area of potential investigation is application of instructional design theory to degree project supervision.

1. Introduction

As stated in the Högskoleförordning [Higher Education Ordinance] (SFS 1993: 100), Educational sciences (henceforth EdScies) are the discipline in which student teachers in Sweden gain their professional degree. In the same educational policy document, it is also stated that the body of studies and work in the profession should be grounded in proven experience and have a disciplinary foundation (Ibid.). Interestingly, as a technical term or discipline, EdScies, in Swedish 'utbildningsvetenskap' are new. Mentioned in official documents for the first time in the early 2000s, the new concept connected teacher education and research to several sectors of the university. Notably, definitions of EdScies range from 'key concept', to 'interdisciplinary field', including for instance educational economics research, historical research on education, and research in the neurosciences (Fransson & Lundgren, 2003), as it is a vast field that offers many opportunities.

2. Background

Action research (2.1), task-based language teaching (2.2) and academic literacies (2.3) form the theoretical background of our study. Thus, in the following, these theoretical approaches along with selected empirical studies will be accounted for. Thereafter, the aim of the study and the research questions will be presented (2.4).

2.1. Action research

Action research (henceforth AR) constitutes both research and practice (Burns, 2010). It is a methodology that has evolved since the mid-1940s aiming to make sense of and build knowledge of the complexities of practice (McAteer, 2018.). The investigation takes place through a combination of different qualitative data collection methods, and it develops and evolves in cycles of research, where the researcher's

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interpretation of data influences the upcoming teaching with regard to the particular aspect in focus (Burns, 2010; Denscombe, 2018). The systematic approach and reflective practice form the foundation for AR projects (Burns, 2010). Through introspection and documentation over a longer period of time, a project gains solid ground (Denscombe, 2018). Despite the criticism AR has received regarding interest in enhancement of practice only, the emancipatory nature of the research activity cannot be overlooked (McAteer, 2018), so participatory AR strives to achieve both (Denscombe, 2018). In this regard, AR is different from daily teaching practice. In the daily work of the teacher, there are seldom affordances for systematic documentation of learning and teaching for research purposes. AR strategies make this possible (Denscombe, 2018). However, also limitations of what is possible to achieve must be considered (Ibid.).

As recently claimed by teacher educators DiLucchio and Leaman (2022), teacher research could be supportive to teachers in many areas of their work. It is supportive to teacher confidence in that there is a methodology to rely on, and that results and analytical frameworks can be shared, and critically discussed with colleagues, locally and internationally. Indeed, the knowledge gained should be discussed also with colleagues in similar contexts elsewhere (Hammersley, 2013). According to DiLucchio and Leaman (2022), teacher education should initiate, support and take interest in AR at schools. AR contributes to building a much needed "body of knowledge about teaching" (Newton & Burgess, 2008, p. 27). Participatory research, i.e. that practitioners themselves construct the research, as well as the ongoing critical discussion among the researching teachers and their context, are crucial ingredients in AR (Hammersley, 2013).

2.2. Task-based language teaching

Task-Based Language Teaching (TBLT) has been developed within the language learning and teaching paradigm, or Second Language Acquisition (SLA) since the late 1970s. TBLT could be perceived as discipline-specific and non-transferable to other contexts of learning and teaching. In contrast, the authors found TBLT to be relevant to interpreting and reporting the results of this study. Therefore, in this study, the student thesis or degree project was considered to be a task.

A task constitutes a unit of analysis assisting language teachers in planning, implementing and evaluating communicative language teaching and learning (Long, 2016). Typically, and importantly, focus on meaning has higher status than focus on form (Ibid.) Similarly, Ellis defines a task as a "workplan that requires learners to process language pragmatically in order to achieve an outcome that can be evaluated in terms of whether the correct or appropriate propositional content has been conveyed" (Ellis, 2003, p. 16). In the achievement of a task, it is necessary for the learner to "give primary priority to meaning and to make use of their own linguistic resources, although the design of the task may predispose them to choose particular forms" (Ibid.). Tasks can be minor assignments that pave the way for a larger task. Moreover, Tasks can embrace considerable content and be highly cognitively challenging (Ellis, 2003).

More recently, Robinson argues in favour of pedagogic task-sequencing where there is movement from less complex to more complex tasks (Robinson, 2022). According this instructional design theory, syllabus design refers to the organisation of tasks as logically linked to purposes of achievement at the overarching level of study, decision-making being made by the teachers and learners (Ibid.) Alignment in task design is advocated also by Skehan; "pre-task activities can have a significant impact on the ways that tasks are done" (Skehan, 2018, p. 132). Mid-tasks, and post-task activities are of relevance for student achievement as well. Planning of tasks should be given more attention. If we would learn and know more about planning of tasks, in the long term-perspective, learning would be easier for students (Skehan, 2018).

According to Bygate (2016), the goal of task-based language teaching is to prepare the learners for the real-world: tasks should be authentic. Ideally, there should be a whole range of empirical studies on different types of tasks that were adapted to different learning environments. This could be useful materials to new and experienced teachers (Ibid.). Ellis emphasizes that student agency in task development has become increasingly interesting, and requested (Ellis, 2003). In school-based research, Wiliam advocates learner agency in design processes: for successful learning, learners should be activated as engineers of their own learning (Wiliam, 2019). Here, in a similar vein, as found by researchers in higher education contexts, students' pre-knowledge plays an important role (Blåsjö & Wittek, 2017; Samuda, 2005; Wendt, 2021).

Task design includes both the development of new tasks and adaptation of existing materials to the current context (Samuda, 2005). According to Samuda (2005, p. 230), surprisingly, "task design remains relatively underexplored as an area of empirical inquiry". As claimed by Samuda, efforts need to be made to "create a window on what task designers really do" (Ibid.).

Task designer thinking and practice was studied in a Swedish school context (Åberg, 2015). In ongoing projects in natural sciences, teachers' and learners' interactive practices with regard to task instruction and learner progression were targeted (Ibid.). Interestingly, it was found that teachers' approaches included attention to learners' pre-knowledge, as well as encouragement of problem-solving and interpretation skills. The teachers proposed questions to the learners, encouraged learners to probe deeper, and challenged learners to make connections with content area previously studied in natural sciences, or with content area studied in other school subjects. Moreover, subtle and often difficult processes of decision-making among the teachers were observed. For example, the teachers were struggling not to provide a direct answer to the learners' questions. Instead, they wanted to encourage the learners to work on finding the answer, together, in their project groups.

Similarly, lecturers supervising student thesis work at the university may be struggling not to give the answer away easily, but instead trying to encourage the student to consider different resources and formulate the answer themselves (Erixon & Josephson, 2017). A requirement, however, is that the student knows what to achieve, that they understand and can respond to the construction of the task (Bygate, 2016; Northedge & MacArthur, 2008). Also, knowing about the rationale for supervisors' scaffolding strategies could be helpful (Nurie, 2018). In this context, Lillis and Tuck (2016) remind us to problematise the notion of instruction as transformation of knowledge in a one-way mode. Rather, it is through negotiation and dialogue that learning and teaching will be successful, processes that "need to be expanded to include multimodality, multi- and translingualism, vernacular and official practices" (Lillis & Tuck, 2016, p. 39).

2.3. Academic literacies

Academic literacies are the 'generic' practices a novice must learn to function in an academic setting (Lea, 2016). This includes fluency in the reading of the texts of the various disciplines and fluency in constructing and communicating own texts within the field (Moje, 2008). More specifically, disciplinary literacy refers to functional knowledge of genres and practices related to genres that are developed by certain groups of researchers, learners and teachers within a specific field (Airey, 2013; Blåsjö & Wittek, 2017; Lillis & Scott, 2015; Shanahan & Shanahan, 2012). Furthermore, adopting an academic literacies approach means recognising academia as diverse, its disciplines and its students (Lea, 2016; Lea & Street, 1998; Lillis & Tuck, 2016).

Literacy approaches need to be integrated (Wingate, 2006). They need to become "an essential part of disciplinary practice, rather than a set of strategies or tools brought in to the disciplines to improve reading and writing of subject-matter texts" (Moje, 2018, p. 99). Indeed, in processes of crossing disciplinary borders, "conventions of one discipline cannot simply be grafted onto another" (Acker, Hill, & Black, 1994, p. 486). Pedagogies need to be explicit, inquiry-based and

context-sensitive (Wingate, 2018).

In cases of transfers between academic disciplines, students' pre-knowledge is a given starting point (Samuda, 2005). Moreover, students' pre-knowledge can function as a source of comparison (Kreber, 2008; Wendt, 2021). By drawing attention to similarities and differences between the academic genres and disciplines that are already known to the students, and the content, procedures and language of the new academic discipline, students' conceptualisation of the disciplines and their genres can develop. As shown in recent research in schools, for development of genre knowledge within and between school subjects, pupils can be activated as resources for their own learning (Berggren, 2013; Wiliam, 2019, 2019berg, 2015).

Both active engagement with the field, and knowledge about the field in question is necessary. Indeed, to gain discipline-specific knowledge, students need to "engage in substantive content-embedded experiences over time" (Langer, 2011, p. 16). Langer (Ibid., p. 17) refers to these processes of sense-making as envision building. Even though Langer's research has been constructed in a school context, with Year 8 learners and teachers of different school subjects, it can be of relevance to higher education. It is relevant because it underscores the importance of learners' active engagement with relevant tasks over a longer period of time. A metacognitive approach is also advocated by Bygate (2016). According to Bygate, newcomers to an academic field will not only need to learn from doing the required assignments, but they will also benefit from knowledge about the discipline. Interestingly, genre pedagogies attend to both these learning processes (Hyland, 2007, 2008, 2013; Lampi & Reynolds, 2018).

Students need to be exposed to a range of studies within the field, discussing in what sense these would be interesting and relevant to the field, and to what extent they could serve as role models or sources of inspiration for students' own upcoming thesis writing. As shown by Råde (2014), these processes are complicated. As regards student thesis writing in EdScies, there seems to be two possible ways of developing clarity: one alternative is that one domain is allowed to dominate, whereas the other alternative is to develop common ground among participants. Råde (2014) argues in favour of the second alternative. A starting point can be to study contexts where there is interest in the matter, an ongoing discussion, and some kind of growing consensus. The Master's thesis context of Pilcher (2011) is mentioned as an interesting resource. In this context, there is open-mindedness and a willingness to adaptation as regards the topics and methodologies of students' research work (Pilcher 2011; in Råde, 2014, p. 31). Carlgren's (2009) study shows that not all supervisors have a clear idea of the aim and boundaries of the student thesis they are going to supervise. The reason for this may be that they are not lecturers with a doctoral degree, that they are not usually directly involved in teacher education, or that they supervise a degree project on a topic that is not within their field or research (Carlgren, 2009; in Råde, 2014, p. 31). Further studies on student thesis writing in higher education in Sweden do not only demonstrate similar findings but also more specific findings. Supervisors need knowledge about writing development at the 'meso level', i.e. paragraphing, change of perspectives, moves from description to interpretation and comparative approaches (Erixon & Josephson, 2017). Notably, two 'genre norms' are identified as significant of student theses in teacher education/EdScies: *relation between own teaching and literature in the field* and *professional statement* (Ibid, p. 233, authors' translation). Supervisors of thesis writing in EdScies should have to learn more about these elements (Ibid.).

Supervision of thesis writing may vary with academic discipline. Whereas a technical rationality model could be expected in the sciences, negotiation would be more common in the social sciences (Acker et al., 1994, p. 486). In fields that reflect mixed traditions of other disciplines, consensus among supervisors might be less common (Ibid.). As regards supervision of student thesis in EdScies, a fairly recent academic field, it could be expected that there is little consensus among supervisors (cf. Erixon & Josephson, 2017). To enhance knowledge of genres, and assessment criteria within various disciplines, students could be encouraged to interview faculty about principles and practices of supervision, or faculty could collect course documents from other courses than those they teach to be discussed in teacher teams (Lampi & Reynolds, 2018). To enhance the situation for writing students, efforts for developing writing instruction (Hirvela & Belcher, 2022), and collaboration among teachers (Wingate, 2018) could also be made. Moreover, research findings on students' views of supervisors' spoken and written feedback on student thesis writing could be considered an essential area to be included Eklund, Mestad, Aksland, & Jegstad, 2022; Neupane Bastola & Hu, 2021; Nurie, 2018; Zhang & Hyland, 2021).

Yet a further perspective on academic literacies are the linguistically, and culturally sensitive pedagogies advocated by Canagarajah (2016). Different from approaches where teachers are "armed with predefined norms, materials, and knowledge for classroom purposes" (Ibid., p. 266), in translingual writing approaches teachers need to continuously "develop their pedagogic knowledge and practice for changing contexts of writing" (Ibid.). In a writing composition course taught by this researcher, course participants work on their literacy autobiography (LA) during a university semester. In this text they integrate reflections on their own planning and teaching of lessons with critical reflections on their own writing development. 'Dialogicality' is the mode through which peers and instructor interact. In recurring feedback sessions, the student writers "have to consider how to communicate their diverse language resources to those who may not understand them" (Canagarajah, 2016, p. 267). Interestingly, in these processes, the student writers "adopt creative strategies to provide clues for readers to guess their meaning" and as readers, the students "develop useful strategies to interpret translingual writing" (Ibid.). Indeed, these interactive processes of negotiation of meaning, seem to contribute to participants' widening their repertoires for interpretation, as well as to developing in-depth approaches to own communication in the written mode.

2.4. Aim of study and research questions

As previously mentioned, action research (AR) enables practitioners to research their own practice (Burns, 2010). It can serve the purpose of enhancing teaching and learning, as well as developing concepts for description and analysis of professional work (Ibid.). There is not only a need for AR in education (Newton & Burgess, 2008), but there is also a need for critical discussion among AR researchers in education (Hammersley, 2013; McAteer, 2018).

Student thesis writing, supervision and writing development in higher education have been researched world-wide (e.g. Ferris, 2007; Hirvela & Belcher, 2007; Nurie, 2018), at times in relation to academic literacies (Lea, 2016; Lea & Street, 1998), disciplinary literacy (Blåsjö, 2004) and genre (Hyland, 2008, 2013). Nevertheless, as regards research on student thesis writing in EdScies, it is still a new and emerging field (Eklund et al., 2022; Erixon & Josephson, 2017; Witsø Veie, 2022).

The study presented here is a contribution in this direction, and it consists of a small-scale AR case study that aims to gain understanding of students' and lecturers' perspectives of the degree project to improve conditions for students' achievement, and to enhance degree project design in the programme. Our research questions are:

RQ1: What conceptualisations of the degree project are outlined?

RQ2: What ideas for future degree project design are developed?

3. Methods

In this section, participants and context (3.1) and material (3.2) are presented, followed by an account of data collection and analysis (3.3), ethical considerations (3.4), methods discussion (5.5) and limitations to the study (3.6).

3.1. Participants and context

The participants consisted of eight students and lecturers in the Bridging Teacher Education Programme (BTEP). The students had recently completed their final term of their studies in the programme. They had received the grades for their student theses, and were waiting to receive their diplomas. The lecturers were coordinators, or teachers of courses in the BTEP. They were working at different departments of the university. The authors of the study, in this context the researchers, were also course coordinators and lecturers in the BTEP, i.e. colleagues of the participating lecturers. We have a background in university studies of languages and literature.

During the three terms in the Bridging teacher education programme (BTEP), students and lecturers engage with the following three sets of courses: subject-content didactics I-III, pedagogy I-III, teaching practice courses I-III and degree project. An overview of the study programme can be found in Appendix A. The degree project (DP) is framed as a ten-week course, during which students independently and under supervision, formulate and investigate a research question related to teaching and learning in their respective school subject(s). In addition to study of relevant literature and empirical studies in the field, the research-oriented work students undertake most often involves data collection, along with analysis of classroom observations, interviews, study materials and curriculum documents with often a combination of two or more sources of data. The outcome of the thesis study is documented, presented, discussed and assessed at the end of the course, in written and spoken format.

3.2. Material

The data material consists of six participatory inquiry conversations with altogether ten participants. The conversations were constructed in the digital meeting room in Zoom and were recorded as audio files. The recorded material encompasses 4.5 h (287.31 min). It was transcribed verbatim in Word (Bryman, 2018; Denscombe, 2018). The written material includes 130 pages of text (57 000 words). The conversations were constructed in Swedish. Transcriptions and translations into English have been made by the authors.

Participatory inquiry with students consists of three conversations with altogether four students. The conversations were constructed in January and February 2021. The data amounts to a total of 118.83 min, or approximately 2 h (see Table 1).

Participatory inquiry with lecturers consists of three conversations with altogether four teachers. The conversations were constructed in March 2021. The data amounts to a total of 168.48 min, or approximately 2.5 h (see Table 2). The researchers, R1 and R2, were present in all six conversations.

3.3. Data collection and analysis

Participatory inquiry was the method we used to study BTEP students' and lecturers' experiences and views of the DP. The participatory conversations were digital meetings in Zoom. We used a conversation guide with a set of questions for the participants to engage with prior to the meeting (see Appendix B). These questions functioned as a structure known to all participants beforehand, and they were occasionally explicitly attended to during the conversations. A graphic overview of participants and topic can be found in Fig. 1. It demonstrates the composition of participants, i.e. the presence of two researchers and 1–2 student(s)/lecturer(s) and the topic of inquiry in focus.

Table 1
Overview of participatory inquiry with students.

Participants	Time	Duration
S1	January 2021	31.26 min
S2 & S3	February 2021	50.51 min
S4	February 2021	37.06 min
= 4 students	= 3 conversations	= 118.83 min

Table 2
Overview of participatory inquiry with lecturers.

Participants	Time	Duration
L1 & L2	March 2021	59.22 min
L3	March 2021	60.02 min
L4	March 2021	49.24 min
= 4 lecturers	= 3 conversations	= 168.48 min

What distinguishes participatory inquiry from the notion of community of inquiry as developed by Garrison (2015) is the composition of participants. Whereas a community of inquiry has its recurring participants, participatory inquiry conversations could have different constellations of participants, depending on the aim and focus of the inquiry. Regardless of changes in aim and focus, sense-making, knowledge-building and innovation are considered to be essential components of participatory inquiry. In several ways, participatory inquiry is similar to focus group interviews (Bryman, 2018; Denscombe, 2018). In an AR project, however, all of the participants are directly involved in the context of the study.

The data was collected during a period of three months, January–March 2021. Before the conversations took place, students were contacted via a message on the course sites of the degree project courses they had recently completed. Course coordinators or lecturers in the programme functioned as contact persons (Bryman, 2018). Course coordinators and lecturers were contacted via e-mail with an invitation to actively participate in the AR study.

Having expressed their interest in participating in the study, an e-mail with further information on the study was sent to the students and lecturers. It included a Letter of Information and a Letter of Consent. The Letter of Information informed on the aim and method of the study, and it included a conversation guide with a set of questions for the participants to engage with prior to the meeting (see Appendix B). In the Letter of Consent, participants agreed to participate in the research project. The Letter of Consent was signed by the participants before the conversations took place (Denscombe, 2018).

Data analysis took place in sequences, during the period of approximately a year. The conversation material was interpreted and analysed by the authors, in collaboration and independently. The method of analysis was qualitative content analysis, constantly comparing data within and between conversations, in relation to the entire material, and in constant search of suitable theory (Ibid.).

3.4. Ethical considerations

In information to the participants, we made efforts to be transparent with the aim and procedure of the AR study, and what it could potentially achieve. We informed the participants that they could contact us, and choose to opt out of the study, at any time, would they find this necessary. To ensure anonymity of the participating students and lecturers, pseudonyms have been used. To further cater for individual

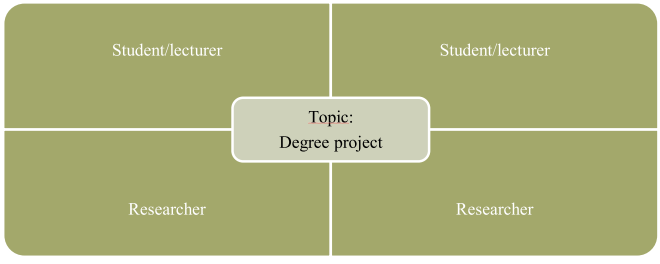


Fig. 1. Participatory inquiry – participants and topic.

participants' integrity in terms of gender, the pronouns 'they' and 'their' have been used. These practices are in accordance with recommendations for good research practice published by the Swedish Research Council (Swedish Research Council, 2017).

3.5. Methods discussion

In retrospect, participatory inquiry conversations could have been combined with other sources of data, for example thinking aloud protocols or student/lecturer log books (see Barnard & Burns, 2012). Another approach would have been for us authors to have been more active as participants in the conversations, and not only being present, encouraging the participants to have their say. At the same time, as an action researcher it is challenging to both be the researcher and a representative of the teaching staff (cf. Burns, 2010). Therefore, in this study, the authors chose to primarily take the researcher role, initiating the conversation, at times summarising points that had been made, prompting further elaboration on the topics, and keeping track of the ongoing recording, and the time taken. A further alternative, would have been to include more participatory inquiry conversations. However, considering our own workload as active lecturers we settled for six conversations with eight participants. Indeed, to cater for quality in an AR project the scope of the collection of data should be realistic (Denscombe, 2018). Hopefully, the small-scale study can contribute to further studies in this teaching and learning context, or provide inspiration to similar studies elsewhere.

In AR, data are constructed to provide perspectives on a certain aspect of teaching and learning, for example by means of new frameworks and methods. These perspectives can contribute to transforming current conditions and pave the way for more effective approaches to teaching and learning (Burns, 2010). Admittedly, there are limitations to what AR can achieve. Changing practices takes time, and in the transformative process, other conditions might be changing, too. As pointed out by Berliner, 2002, researcher in the field of EdScies: as we are building the bridge, the landscape is simultaneously moving.

Throughout the research process, we have tried to work with an open mind. Throughout the analysis procedure we have made efforts to work in a trustworthy way. The voices of the participants should be heard (Barnard & Burns, 2012). That is first priority, to be honest toward the data material. Then again, we cannot help that we are the researchers, teachers, previous students, and current authors that we are. It could be that our language is not as expressive or concise as the language of the inquiry-oriented participants were there and then, in the real conversations. We did our utmost to make the material speak for itself, in a research article mode.

As teacher educators researching the field of EdScies, we must also attempt to view the field from an exterior perspective, attempting to conceptualise the DP at the university-wide and societal level. If this is not the case, the function of the DP as a means of professional development cannot be justified. If we cannot incorporate the etic perspective as well, our critique cannot be considered solid or robust (Berglund, 2016). In this respect, we have engaged with research literature from a range of academic disciplines, in Sweden and abroad, closely linked to teacher education (e.g. Blåsjö & Wittek, 2017; Canagarajah, 2016) or not directly linked to teacher education, e.g. student learning in engineering (Entwistle, 2009; Scheja, 2002), and student writing in economics and history (Blåsjö, 2004), as well as with empirical studies from other educational contexts than higher education, e.g. school contexts (Berggren, 2013; Langer, 2011, 2011berg, 2015). Policy documents published by Swedish education authorities (SFS, 1993, p. 100; Fransson & Lundgren, 2003) have been included, as have critique of reforms (Beach, 2011). These readings have revealed to the researchers both possibilities and challenges with EdScies as the discipline in which teacher students in the BTEP achieve their final degree project.

3.6. Limitations to the study

Limitations to this AR study are that findings are closely integrated with their context; student thesis writing in the BTEP at a university in Sweden. Thus, they are not generalizable to other contexts. There are also limitations to the study as regards the knowledge and experience of the authors. Being researchers with a background in language education, we have constructed the study out of our knowledge-base and view of the matter.

4. Results and analysis

This section presents results and analysis. Two main themes emerged from the analysis of the participant data: conceptualisation of the DP and suggestions for improvement. In the following, these themes will be introduced and explained.

4.1. Conceptualisation of the DP

In terms of participants' conceptualisation of the DP three areas emerged: scepticism toward the DP, disciplinary boundary crossing and time management. Scepticism toward the DP was commonly articulated by the students, whereas disciplinary boundary crossing and time management in the study programme were mentioned by both students and lecturers. For an overview, see Fig. 2.

Student S4, future upper secondary school teacher of history, expressed scepticism toward the DP. Even though this student had recently completed their Master's thesis in the field of the humanities; history, the student thesis in EdScies was difficult to conceptualise. The thesis genre, its aim, method and language (see Hyland, 2008), could not easily be conceived by the student. The student teacher comments on their recent writing achievement with a combination of surprise and frustration:

> To my view, it became a form-focused task. Much more rigid in some way … That the final product became a report … It was not something I was prepared for. (S4)

Student, S1, a future upper secondary school teacher of languages, who had recently completed their Master's thesis in the field of the humanities; languages and literature, was also surprised at the extent to which focus on form played a role in the completion of the student thesis in EdScies:

> During my previous thesis work [in the field of the humanities], focus was very much on the content. You studied a content area in-depth. Often it had to do with a literary work. You consulted hand books … You were supposed to stick to a certain form, but this form was looser … [and not as strict as the format for the student thesis in educational sciences]. (S1)

With a touch of scepticism and frustration, S1 comments on the thesis examination seminar at the end of the course:

> The content [of our theses] we did not … there was hardly any room for it at all. Talking about the final seminar, we did not mention [the content of the thesis work] at all. Instead, everything had to do with

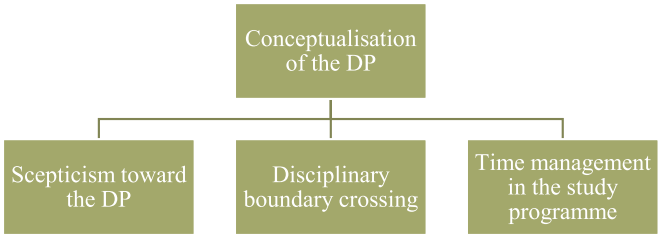


Fig. 2. Overview of participants' conceptualisation of the DP.

> the form. I don't know. I was not prepared for this [unfolding of the event], really. (S1)

As shown, students S4 and S1 expressed scepticism toward the DP in several aspects; how it was introduced to them, how their work with the student thesis unfolded, and how the thesis was presented and assessed in the examination event. As claimed by both students, at their (different) final examination seminars, there was no, or only limited focus on content knowledge gained from working with the thesis, only issues and comments that related to formal aspects of the thesis text. From a TBLT point of view, this is problematic. A task should be meaningful. Its main focus should be on the content (Ellis, 2003; Long, 2016). Moreover, a task should be authentic (Bygate, 2016). Interestingly, the students who had recently completed their Master's degrees within the humanities, languages and literature and history, respectively, were troubled by the extensive focus on form throughout the DP process, at the expense of content. What constitutes a student thesis in EdScies? What are the possibilities and limitations? These are questions of potential interest and relevance to lecturers and students involved in the emerging field of EdScies.

Student S3, a future teacher of mathematics, was also struggling with conceptualisation of the DP. S3's experiences of difficulty relate to the areas disciplinary boundary crossing and time management:

> I can say that science, or educational sciences, it is research. We are not all researchers. And there was so much, much, much … I can say that there was much to do in a very short time period. You know, we were supposed to study articles, books, those things. For example, myself I could not manage with only studying four articles and two-three chapters of the book. And I do think that one has to be a little bit more … Perhaps I should put it like this … more explicit, science with a focus, and not jump to and fro, what does this [researcher] say, and what does this other [researcher] say, compare. It was too much for me, you know, I have another language as well. (S3)

First, S3 is bothered by the encounter with the student thesis genre; the aim, methods and scope of the DP, as it has been perceived. It does not make sense to the student: it is so different from their previous experience of thesis writing. Secondly, S3 is troubled by the time management in the programme. Indeed, student S3 cannot make sense of the logic and relevance of this DP in the light of the short time during which the thesis is going to be studied and completed, i.e. ten weeks. The rationales for the DP, why the students should be posted as researchers, and why this should be in the field of EdScies, has not become clear to the student, not even at the point of the participatory inquiry, which takes place after completion of the thesis work. S3's experiences resonate with Norwegian teacher students' feelings of uncertainty with regard to their research assignment (Eklund et al., 2022).

More specifically, students' experiences of difficulty related to time management entails two notion. First of all, it relates to time management in the BTEP. As mentioned by student S3, the relation between the huge scope of the DP and the limited time during which it is going to be achieved is problematic. Secondly, the theme refers to time management regarding the amount of time that may have elapsed between students' previous academic studies, and the enrolment in the BTEP. For example, a fellow student of S3's, S2, future teacher of mathematics in upper secondary school. For this student several decades had passed since their studies in engineering, and the time during which the student produced thesis work for their Civil Engineering Degree. As mentioned by students S2 and S3, this time span could be catered for in the study programme, in combination with attention to, and acknowledgement of the work experience, and life experience, these students bring to the BTEP (see further 4.2).

Turning now to the lecturer perspectives, the lecturers expressed their awareness of the students' difficulties in navigating the genres of EdScies. For example, on several occasions in the participatory inquiry, the theme of disciplinary boundary crossing was addressed and interpreted:

> They [the students] are basically lost in a new research tradition! … They [the students] cannot navigate in the landscape of educational sciences. … They do not conceptualise social sciences methodology as research … They [the students] have come from other teaching and learning- and research traditions (L1).

When there are difficulties in the supervision process, it is often because we have different understandings of what kinds of texts that are going to be produced (L3).

As pointed out by lecturer L1, student teachers may not conceptualise research in the social sciences as research, especially not if they have their academic foundation in the disciplines of mathematics or natural sciences. This reasoning is in line with insights and recommendations in academic literacies. Academic literacy in a 'generic sense' is not the same as disciplinary literacy (cf. Acker et al., 1994; Airey, 2013; Blåsjö & Wittek, 2017; Shanahan & Shanahan, 2012). Instead, active engagement with the discipline and its genres need to be the case (Lillis & Tuck, 2016), managing DPs over a longer period of time (Robinson, 2022) and building metacognitive knowledge (Bygate, 2016).

As regards thesis supervision, lecturer L3 addresses the need of attending to expectations of the genres of student thesis texts (cf. Acker et al., 1994; Bygate, 2016; Moje, 2008; Wingate, 2006). Becoming literate in the field does not only take time and active student engagement, it also requires a dialogical approach (Lillis & Tuck, 2016; Wingate, 2018). It may also require socially, culturally and linguistically sensitive pedagogies (Canagarajah, 2016).

Interestingly, in one of the participatory inquiry conversations with the lecturers, a further aspect of disciplinary boundary crossing is addressed, namely students' professional development:

> L2: Perhaps we should view this as a culture clash from their [the students'] points of view about what it is, becoming a teacher.
>
> R2 & R1: Mm, yes.
>
> L1: Yes, that's a way of looking at it!

In this extract, the multi-faceted dimension of the teaching profession is highlighted. Lecturer L2 hints at the complexity of the profession, and the complexity embedded in conceptualising "what it is, becoming a teacher". L2 gives an indication of the many components and perspectives students need to integrate into their professional selves during their time in pre-service education. Indeed, as stated in the Higher Education Ordinance (SFS 1993:100), professional work should be permeated by proven experience and build on scientific evidence. Becoming a teacher takes time. It is a complex, multidimensional profession. Therefore, it could be argued that the vast field of EdScies is what is needed. Potentially, the student thesis provides a space for exploring issues related to teaching and learning in their school subjects that student teachers may have. However, the vast field with its immense opportunities requires knowledgeable supervisors (Carlgren, 2009; In: Råde, 2014), something which calls for further knowledge-building and collaborative research and teaching.

If the "culture clash" mentioned by lecturer L2 would be considered as a prompt to develop DP design in the BTEP, academic literacies should be addressed throughout the programme. If disciplinary boundary crossing would be noted as a prompt to DP design in the BTEP, implications are to construct pre-DP activity (cf. Skehan, 2018) that prepare the students for the pending thesis work, and indeed for the profession. Metacognitive approaches, as advocated by Bygate (2016) would be one means of facilitating students' navigation between disciplines. By constructing tasks that prepare students for thesis work, and by making students aware of the sequencing of tasks in the programme as stepping stones to their becoming teachers, they would both achieve better and feel better about their achievement. This is requested by all four students in this study. According to the students, if they had known more of the aim, procedure and language of the student thesis genre at

an earlier point in the BTEP, they would not have experienced feelings of scepticism, frustration and difficulty to the extent that they did.

In addition to students' mentioning of time management as problematic, it is commented on in the lecturer conversations. The short duration of study in EdScies is addressed, together with considerations on the need of preparing students for the thesis DP throughout the set of courses in the programme:

> It is a short programme! It is very important what kind of preparations that take place. The meta-level understanding – while engaging with research in teaching and learning, or in other research in the field of educational sciences, [that we] also talk about the methods, or at least mention the methods, you know. (L2)

Similarly, lecturer L4 underscores challenges related to development of disciplinary literacy within EdScies, given the intensity of the programme. During a three-term programme students face courses in the teaching and learning of the school subjects, or subject-content didactics, followed by courses in pedagogy and teaching practice at schools. To this teacher's knowledge, pre-task efforts are continuously made in the courses preceding the thesis writing course. For example, in the subject-content didactics courses, types of research questions in the articles studied are highlighted as potential research questions for a student thesis. Furthermore, a coming project was also mentioned by lecturer L4, where students would have the opportunity to study the final school practicum course, and work with the DP in parallel.

4.2. Suggestions for improvement of DP design

In the participatory inquiry, students and lecturers outlined and developed many ideas for improvement of the DP design. In most cases, these suggestions originated in their own experiences of learning and teaching, respectively. In some cases, the conversations served as meaning-making events, where new ideas came into being as the participants were interacting in the digital meeting room. Three major themes emerged: preparatory course, step-by-step procedure and metacognitive approach. Both learners and teachers tapped equally into the three areas. For an overview, see Fig. 3.

Student S1 speaks in favour of a preparatory course at the beginning of the BTEP:

> Perhaps there could have been some kind of preparatory course, some kind of preparations for the research-oriented DP. This is my opinion. (S1)

Student S4 is also speaking in favour of early preparation for the thesis writing project. In doing this, not too surprisingly, really, the student teacher is placing themselves in the position of the teacher:

> [I]f I would have been the teacher or supervisor. Then I would have had an introductory conversation with my students: 'This is the framework when you study for your teacher education thesis at our department. It is important to think about these areas.' (S4)

Having found the organisation of the thesis writing course at the department where the previous thesis was accomplished to be effective, S4 makes suggestions for a similar DP design in the context of the thesis in teacher education. There should be a clear introduction to thesis writing, and students should be requested to submit a thesis proposal prior to embarking on the course.

> There should be some kind of more explicit introduction. We didn't submit a thesis proposal. There wasn't anything like that. The practices were more that we shared work in progress now and again. (S4)

> A short intro. Alternatively, a compulsory submission of a thesis proposal. I have done that before, during my previous studies. It was really worthwhile. (S4)

In this conversation, student S4 is tapping into modes of explicit-making the discipline and the genre (e.g. Hyland, 2008), addressing students' previous knowledge (Samuda, 2005), considering task-sequencing (Skehan, 2018) and metacognition (Bygate, 2016) (see further 2.2). S4 has a clear idea of how the student thesis could be introduced to students at the start of the course. Drawing on experience from a similar course in another academic discipline, the student is using comparison as a means of analysis and reflection (Entwistle, 2009; Kreber, 2008; Wendt, 2021).

Students S2 and S3, future teachers of mathematics, pick up on the same topic, extending it to reasoning about a step-by-step procedure. To these students, this would be a design for successful DP achievement. Here, they include the teaching practice courses as potential components in the DP design (for overview of the BTEP, see Appendix A). Their suggestions range from information about the degree project at the very start of the BTEP, to formulation of early thesis proposal after the first 5-week teaching practice course, to having the opportunity to get started with the thesis project at the time of the second 5-week teaching practice course.

> S3: Yes, exactly. As I said earlier, it is better for us to get started even before the first school placement course. Or, that you come and give us information in a clear way.
>
> R1: Mm.
>
> S3: But we could, we could submit … after the first school placement course, we could submit our project plan. And then we would have plenty of time to consider how to formulate ourselves for receiving a passing grade.
>
> R1: This could be an option, couldn't it? Perhaps not everybody has to study the course this way.
>
> S3: Yes, exactly. / … /Alternatively, there could be, I don't know, it could start at the same time as the third and final school placement course. They could be integrated, Teaching practice 3 and, what's it called, the Student thesis writing course.

To student S3, the step-by-step procedure allows for cognitive development over a longer period of time:

> To my view, there is so much more information that could be provided before we start the first school placement course. You [the lecturers] could tell us that we could start thinking [about the outline of the project] This … Of course, At the beginning of the student thesis writing course we received a document where we had to decide on topic for the student thesis. We were to choose what to write about. In my opinion, this document should have been sent to us at the start of the study programme. That would not have been so late. (S3)

Student S3 is emphasising the need for students to be given more time to develop their thinking. To this student, the intensive process of receiving information, having to choose topic and getting started with data collection, then analysing the data and writing the research report in only ten weeks is too stressful. In comparison, S3's reasoning resonates with the notion of task-sequencing in language teaching, as

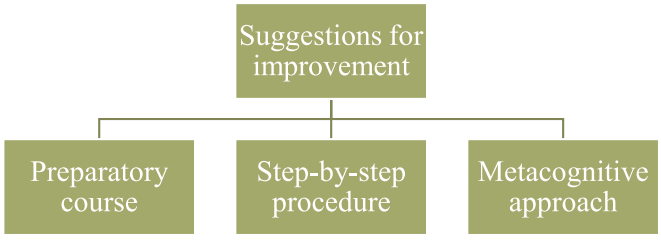


Fig. 3. Overview of suggestions for improvement.

studied by Robinson (2022). Within this discipline, planning for movement from less complex tasks to more cognitively challenging tasks has proved to be conducive to learning (Ibid.).

In the conversations with the lecturers, the mention of pre-DP activity is frequent. To the lecturers' knowledge, pre-DP activity, in different forms, is implemented at different stages in the programme (see also 4.1). That there had been plans to prepare students for the DP in a more long-term and organised way, at the programme level, some three years ago, is also mentioned in the conversations with the teachers, but that these plans for some reason got stranded along the way.

In the conversation with students S3 and S2, researcher R1 explicitly engages in a dialogue on improving conditions for students' successful achievement of the DP:

> R1: There have to be systemic efforts made then. Adjustments and improvements should not be made randomly. The structure of the programme could be better. We agree on this. There could be better progression in the study programme. At least, you would have an opportunity to prepare for the project in educational sciences, then, the student thesis, I mean.
>
> S3: Yes, exactly.

In the same conversation, student S2 takes a metacognitive perspective:

> S2: However, according to me, there is an effort that should be made on the part of the university.
>
> R1: Yes.
>
> S2: The supervisors should study a course in supervision of student thesis work so that they act upon the same principles and practices, because they don't.
>
> R1: No. Good suggestion. A set of shared principles and practices is required, cross-faculty.

As regards improvement of supervision of student thesis, researcher R1 and student S2 agree. Supervisors need to develop a set of shared ideas about supervision of student theses in EdScies.

5. Discussion

In this section, we discuss our findings in the light of theory and previous studies. In pursuit of the research questions, RQ1 and RQ2, we discuss participants' conceptualisations of the degree project (5.1) followed by a discussion of participants' suggestions for development of degree project design in the study programme (5.2). A discussion of method can be found in a previous section of the article (3.5). The section ends with a conclusion (5.3).

5.1. Conceptualisations of the DP

Student thesis writers need to know what to expect, and what is expected of them. As shown in this study, students' pre-knowledge should be activated as a potential resource for learning, at the beginning of the thesis writing course, or even earlier in the BTEP. Providing students with time to think and make early preparations for their thesis work be a means of dealing with the scope of the short programme, and a means of being pro-active as regards 'delayed understanding' (Scheja, 2002). Interestingly, student teachers' in the Norwegian teacher education context express similar views of thesis writing as overwhelming and stressful (Eklund et al., 2022). Moreover, as suggested by S2, by viewing the EdScies thesis work in the light of students' previous theses projects, there is an opportunity to learn through comparison, applying a compare- and contrast approach. By addressing students' pre-knowledge (Langer, 2011; Samuda, 2005), and by making students aware of the knowledge they possess in a way that they can also articulate it, using context-sensitive pedagogies (Christensson, 2021; Erixon & Josephson, 2017; Canagarajah, 2016), transitions between disciplines will be less challenging to conceptualise and achieve.

Student teachers, finding themselves in the field of EdScies need to be introduced to a range of different studies that can be undertaken within this vast field, and be provided with aims and methodologies for analysis and reflection. As accounted for by lecturer L4, this is part of the subject-didactics courses in the BTEP. By presenting students with examples of research in the field, and by encouraging students to reflect on the usability of this research for their future teaching, students are prompted to consider the potential applicability of this research perspective of their own pending research-oriented study, i.e. the student thesis. As pointed out by students S2 and S3, teaching practice courses could also contribute to building this knowledge. However, in terms of planning, supervision and assessment, student thesis work integrated with teaching practice courses calls for collaborative DP design between university and schools.

As came to the fore in the dialogue between student S2 and researcher R1, there is lack of consensus among university supervisors of the student thesis. This is not a new situation (Acker et al., 1994; Råde, 2014; Erixon & Josephson, 2017). Educational sciences as the discipline for student teacher thesis projects require structures that need to be developed in collaboration, and shared with students and colleagues within, and outside the field of teacher education for response (see Beach, 2011; Englund, 2006; Råde, 2014). A repertoire of principles and practices for supervision of thesis writing is desirable. Models from other fields than higher education can provide structures and inspiration for teacher education. As regards collaboration and knowledge building in thesis supervision, AR in the BTEP could build on experiences for example from school contexts on partnerships for literacy (Langer, 2011). Findings from these partnerships for literacy projects showed that they were building students', as well as teachers', knowledge about subject-disciplines. They facilitated discussions about, and, so also, transfers between subject disciplines (Ibid.)

5.2. Development of DP design at the programme level

Drawing on language education theory, the authors' claim is that the DP as a key component of the BTEP would benefit from a proactive approach, namely DP-sequencing (cf. Robinson, 2022), where minor tasks pave the way for the major task, here the DP (cf. Ellis, 2003), and where minor challenging tasks are followed by more challenging tasks (Skehan, 2018). Indeed, there could be both student motivation and enhanced learning to be gained from awareness-raising of adjacent courses in the programme, i.e. teaching staff demonstrating that they do not only know their own course in parts and parcel, but that they also can communicate to the students how their course is constructively linked to the other courses in the study programme. Efforts would need to be made in terms of knowledge-building among teachers teaching adjacent courses in the programme. Inspiration can be found in the community of inquiry model (Garrison, 2015). Models for planning and practice are also to be found in other contexts that resemble the BTEP context (e.g. Pilcher, 2011; in Råde, 2014), or in successful literacy developed projects studied and developed in school contexts (Langer, 2011). This is in line with a metacognitive approach as advocated by Bygate (2016) in the field of language education and also speaking the language of constructive alignment as promoted by Biggs & Tang (2011) in higher education at large. Here, our claim is to take on the prompts of task-sequencing, metacognition and constructive alignment for development of DP design at the study programme level.

As noted by lecturers L1 and L2, professional development is a complex process. Studying the degree project course could be perceived as a "culture clash" (L2). There are many factors to take into consideration in trying to understand "what it is, becoming a teacher" (L2). Even though suggestions were made to integrate thesis writing with teaching practice courses, there were also participants who mentioned that the cognitive load could be too heavy, having to focus on teaching and

theory simultaneously (cf. Eklund et al., 2022). Each individual student teacher is unique in their knowledge-base, and, using Langer's (2011) language, in their envisionment building of the profession and its DPs and assignments. This must be taken into consideration, and responded to, e.g. in terms of context-sensitive pedagogies and by means of reflection for learning, through speaking and writing, or other modalities (Canagarajah, 2016; Christensson, 2021). Consequently, a plurality of perspectives needs to be included on a continuous basis. Collaborative design at the programme level is a dynamic event. As found in this study, participatory inquiry conversations are a means of developing collaborative programme design in the BTEP, enhancing student achievement of the DP, both cognitively and emotionally, facilitating students transition between academic disciplines. Last but not least, it would be meet the aims of 'proven experience' and 'disciplinary foundation' that should permeate all work in teacher education (SFS 1993: 100).

5.3. Conclusion

Our conclusion is that conditions for students' successful achievement of the DP in the BTEP could be improved. As demonstrated in this qualitative action research pilot study, students conceptualised the DP in EdScies as unclear, challenging and even unnecessary. This is not ideal situation for teachers to-be. Alignment in syllabus design, including awareness-raising of the DP at earlier stages in the programme together with an academic literacies approach to thesis writing and supervision of thesis are suggestions of improvement that emerged from this study. In combination, they could make a difference for future generations of students, cognitively and emotionally.

We would like to recommend participatory inquiry conversations as a useful method. The aim of the participatory inquiry being to shed light on an issue of interest by bringing in different perspectives on the matter, participants should feel that they are being listened to, and that they contribute to sense-making of this matter and to knowledge-building of the field.

As shown in this study, AR helps us understand our own context, and it can contribute to enhancing the work we do. In a wider context, AR projects can contribute with analytic frameworks and concepts for teaching and learning, e.g. student thesis writing. Transitions between disciplines can become less challenging for student thesis writers and their supervisors.

Implications for further AR research and teaching as regards the DP is teacher collaboration at the study programme level, within and between courses and across teacher teams, including studies of degree project genre instruction.

Credit author statement

Y. Sandberg: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Supervision, Project administration, Writing – original draft preparation, Writing – Reviewing and Editing, Visualisation, B Fröjdendahl: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Supervision, Project administration, Funding acquisition: Not applicable

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Overview of the three-term BTEP

Term 1	**Subject-Content Didactics I**	Pedagogy I	Teaching practice I	Subject-Content Didactics II
Term 2	Pedagogy II	Pedagogy III	Teaching practice II	Subject-Content Didactics III
Term 3	Teaching practice III	Teaching practice III	**Degree Project**	**Degree project**

Appendix B

Preparation for participatory inquiry

The aim of our study is to investigate what you already know about scientific research and the production of academic text; what you would need in terms of such knowledge; as well as what means of preparation you would need during your studies in pre-service teacher education, in particular as regards the teaching practice courses.

Please, find a few questions to consider prior to the conversation in Zoom.

1. What do you already know about the planning and construction of a research assignment in educational sciences? Please, feel free to elaborate on your considerations regarding this question.
2. Give an account of the decision-making of topic regarding your student thesis. How did you make your choice of topic? According to you, in what way is it difficult to limit the scope of the topic?
3. What are your thoughts on the possibilities of planning, and/or construction of data for the student thesis before the first school placement course, or during any of the university-based courses in the BTEP?
4. Do you have any further comments or questions?

(extract from Letter of Information, sent to students prior to the participatory inquiry Zoom conversations).

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